Regression equation of Y on X,

$$\hat{Y} - \bar{y} = b_{YX}\left(x - \bar{x}\right)$$

Calculation of b_{YX}

$$b_{YX} = b_{VU}\frac{n\sum_{i=1}^{n} u_i v_i - \sum_{i=1}^{n} u_i \sum_{i=1}^{n} v_i}{n\sum_{i=1}^{n} u_i^2 - \left(\sum_{i=1}^{n} u_i\right)^2}$$

$$= \frac{5\times 970 - 48\times 9(-12)}{5\times 2094 - (48)^2}$$

$$= \frac{4850 + 576}{10470 - 2304}$$

$$= \frac{5426}{8126} = 0.664$$

$$b_{YX} = b_{VU} = 0.664$$

$$\hat{Y} - 72.6 = 0.664\ (x - 69.6)$$

$$\hat{Y} - 72.6 = 0.64x - 46.214$$

$$\hat{Y} = 0.664x + 26.386$$

Regression equation of X on Y,

$$\hat{X} - \bar{x} = b_{XY}\left(y - \bar{y}\right)$$

Calculation of b_{XY}

$$b_{XY} = b_{UV}\frac{n\sum_{i=1}^{n} u_i v_i - \sum_{i=1}^{n} u_i \sum_{i=1}^{n} v_i}{n\sum_{i=1}^{n} v_i^2 - \left(\sum_{i=1}^{n} v_i\right)^2}$$

$$= \frac{5\times 970 - 48\times(-12)}{5\times 1074 - (-12)^2}$$

$$= \frac{4850+576}{5370-144} = \frac{5426}{5226}$$

$$b_{UV} = 1.038$$

$$\hat{X} - 69.6 = 1.038\ (y - 72.6)$$

$$\hat{X} - 69.6 = 1.038y - 75.359$$

$$\hat{X} = 1.038y - 5.759$$

12th_Statistics_EM_Unit_5.indd 143 1/28/2020 6:36:43 PM